

Juan Pablo Garcia Chavez

Junior Machine Learning | Python | TensorFlow | Scikit-learn | Pipelines

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Portfolio: <https://JuanPa7799.github.io/juanpablo-data-portfolio/>

Profile

Junior Machine Learning practitioner with a background in Mechatronics Engineering and a Master's degree in Electronic Engineering. I work with classification, regression, computer vision, NLP, forecasting, and signal-focused projects. My current focus is building reproducible solutions and moving from notebooks toward reusable workflows.

Skills

- Python, scikit-learn, TensorFlow, Keras, LightGBM.
- Training, validation, and evaluation pipelines.
- Computer vision, NLP, forecasting, and signal processing.
- Trade-off analysis between predictive quality, speed, and complexity.

Selected Projects

Electrical disturbance classification

Flagship ML project applied to electrical signals and power quality.

- Reported accuracy: 96%+.
- Approximate processing reduction: 80%.
- Stack: Python, NumPy, pandas, scikit-learn, TensorFlow.

Age estimation with computer vision

ResNet50 model to estimate age from images.

- Validation MAE: 7.03.
- Architecture: ResNet50 + regression head.
- Responsible approach: operational support, not a single decision maker.

IMDB review sentiment

Text classification with TF-IDF, logistic regression, LightGBM, and BERT test.

- Required F1 threshold: greater than or equal to 0.85.
- Best approach: TF-IDF + logistic regression.
- Focus: practical, explainable, and efficient NLP.

Taxi order forecasting

Temporal pipeline with lags, rolling means, and chronological validation.

- Final model: Random Forest.
- Test RMSE: 43.21.
- Operational target met.

Education

- Master's degree in Electronic Engineering.
- Mechatronics Engineering.
- Applied training in Data Science and Machine Learning.

Links

- Machine Learning dashboard: <https://JuanPa7799.github.io/juanpablo-data-portfolio-/dashboards/machine-learning/>
- Main portfolio: <https://JuanPa7799.github.io/juanpablo-data-portfolio-/>